

CXR ELK Lab Manual

SW.12 — logs (Elasticsearch + Kibana + Filebeat)
Connecting Claim Studio :3000 to Kibana

CXR Ops Lab — `cxr-ops-lab`

2026-05-31

Abstract

Step-by-step lab for **SW.12 ELK**: every file added for log shipping, how the **Claim Studio dashboard** on **:3000** produces Docker logs that Filebeat indexes, and how **Kibana (:5601)** is where you search them. Contrasts with **Jaeger (:16686)** traces (SW.11) and **Grafana (:3001)** metrics (SW.9–10). After you close the lab, ELK may stay stopped; Jaeger remains useful for rehearsal **:8251**.

Contents

1	Where you view telemetry (three pillars)	2
2	How :3000 Claim Studio connects to Kibana (highlight)	3
2.1	End-to-end path	3
2.2	What “connects” the main dashboard	3
2.3	What is <i>not</i> wired (important)	3
2.4	Jaeger vs Elasticsearch (when ELK feels like overkill)	4
3	Step-by-step procedure	5
3.1	Step 1 — Start Compose :3000 (log source)	5
3.2	Step 2 — Start ELK stack	5
3.3	Step 3 — Golden path: Claim Studio → logs → Kibana	5
3.4	Step 4 — Kibana UI tips	5
3.5	Step 5 — Write evidence	6
3.6	Step 6 — Stop ELK (optional, after lab)	6
4	Files created for SW.12 (full inventory)	7
4.1	Files touched (not new)	7
4.2	Related observe stack (SW.9–11, separate compose)	7
5	Port map (CXR lab)	9
6	Implementation checklist (repeat on another machine)	10
7	Troubleshooting	11
8	Study plan linkage	12
9	Live adjustments (Kibana path + shipping)	13
10	UI evidence — Kibana Discover (:5601)	14

Chapter 1

Where you view telemetry (three pillars)

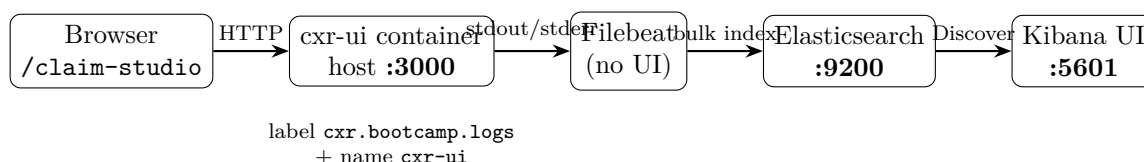
Signal	Tool	URL	Bootcamp SW	Web UI?
Metrics	Prometheus	:9090	SW.9	API
Dashboards	Grafana	:3001	SW.10	Yes
Traces	Jaeger	:16686	SW.11	Yes
Logs	Kibana	:5601	SW.12	Yes
Log store	Elasticsearch	:9200	SW.12	API (no UI)
Shipper	Filebeat	(container)	SW.12	No UI

Main CXR app (Claim Studio): <http://localhost:3000/claim-studio> — this is the **product dashboard** you use to run analysis. It does *not* embed Kibana or Jaeger; those are **separate observability UIs** you open in other browser tabs when debugging.

Chapter 2

How :3000 Claim Studio connects to Kibana (highlight)

2.1 End-to-end path



2.2 What “connects” the main dashboard

1. **Compose CXR UI running** — `./scripts/04-compose-up.sh` starts the `cxr-ui` service on **:3000**.
2. **Label on the service** — `compose.yaml` sets `cxr.bootcamp.logs: "true"` so Filebeat autodiscover selects the container.
3. **ELK stack running** — `./scripts/16-elk-up.sh` starts Elasticsearch, Kibana, Filebeat (`compose.elk.yaml`).
4. **Filebeat config** — `observe/elk/filebeat.yml` watches Docker via socket; matches container name `cxr-ui` or the label above; reads `/var/lib/docker/containers/.../*.log`.
5. **User action on Claim Studio** — page loads and **Run Analysis** write Next.js / server lines to the container log stream.
6. **Indexing** — Filebeat ships JSON events to Elasticsearch (indices like `filebeat-*`).
7. **Kibana** — Data view `filebeat-*`, **Discover**, filter e.g. `claim-studio`; screenshot for evidence.

2.3 What is *not* wired (important)

- **Rehearsal :8251** (`npm run dev:rehearsal`) — logs go to the *host terminal*, not the `cxr-ui` Docker container. SW.12 evidence uses **:3000** Compose traffic unless you add a custom Filebeat input later.
- **Jaeger (SW.11)** — traces from :8251 with `OTEL_*`; does *not* replace log search. Jaeger answers latency; Kibana answers “what did the server print?”

- **Grafana (:3001)** — metrics dashboards; not application log lines.

2.4 Jaeger vs Elasticsearch (when ELK feels like overkill)

For a **single-service** bootcamp app, running ELK every day is optional.

- **Keep for the lab:** prove SW.12 (Discover screenshot + `evidence/SW12-elk-verify-2026-05-29.md`).
- **Then:** `docker compose -f compose.elk.yaml` down to free RAM; keep `07-observe-up.sh` + Jaeger for trace work on :8251.
- **Production pattern:** many teams run logs + traces + metrics; syllabus teaches all three.

Chapter 3

Step-by-step procedure

3.1 Step 1 — Start Compose :3000 (log source)

```
cd cxr-ops-lab
./scripts/04-compose-up.sh
curl -sf http://localhost:3000 >/dev/null && echo "OK :3000"
```

3.2 Step 2 — Start ELK stack

```
./scripts/16-elk-up.sh
./scripts/16-elk-smoke.sh
```

Wait up to ~60s after first start for Kibana to answer on **:5601**.

3.3 Step 3 — Golden path: Claim Studio → logs → Kibana

1. <http://localhost:3000/claim-studio>
2. Click **Run Analysis** (or reload pages several times).
3. Wait ~30s for Filebeat to index new lines.
4. <http://localhost:5601> → **Explore on my own** (skip Add integrations).
5. **Stack Management** → **Data Views** → create `filebeat-*` (timestamp `@timestamp`).
6. **Discover** → select data view (e.g. “new CXR”) → time range **Last 24 hours** → refresh.
7. Optional KQL: `claim-studio` or `analyze`.
8. Screenshot for `evidence/SW12-kibana-discover-*.png`.

3.4 Step 4 — Kibana UI tips

- Ignore **Security** sidebar (Alerts/Findings) on Home — not required for this lab.
- Histogram brush zoom: **undo arrow** only appears right after brushing; otherwise click the **time range** picker → **Last 24 hours** → Update + refresh.
- Expect fields such as `agent.name: filebeat`, `container.id`, `@timestamp`.

3.5 Step 5 — Write evidence

Edit `evidence/SW12-elk-verify-2026-05-29.md`: checkboxes for data view, Discover screenshot, golden path note.

3.6 Step 6 — Stop ELK (optional, after lab)

```
docker compose -f compose.elk.yaml down
```

Claim Studio on :3000 keeps working without ELK.

Chapter 4

Files created for SW.12 (full inventory)

File	Purpose
compose.elk.yaml	Elasticsearch :9200, Kibana :5601, Filebeat; volume elk_es_data
observe/elk/filebeat.yml	Docker autodiscover → cxr-ui / label cxr.bootcamp.logs
scripts/16-elk-up.sh	docker compose -f compose.elk.yaml up -d
scripts/16-elk-smoke.sh	Health: ES, Kibana :5601, Filebeat running, filebeat-* index
scripts/build-elk-manual-pdf.sh	Builds this PDF from CXR-ELK-LAB-MANUAL.tex
docs/CXR-ELK-LAB-MANUAL.md	Markdown twin of this manual
docs/CXR-ELK-LAB-MANUAL.tex	LaTeX source for PDF
docs/CXR-ELK-LAB-MANUAL.pdf	Printable lab record (run build script)
evidence/SW12-elk-verify-2026-05-20.md	Verifies SW.12 + agent smoke notes

4.1 Files touched (not new)

File	Change
compose.yaml	Label cxr.bootcamp.logs: "true" on cxr-ui service
docs/OBSERVE-WIRING.md	SW.12 URLs and commands
docs/PLATFORM-SYLLABUS-MAP.md	Maps SW.12 to ELK compose + manual

4.2 Related observe stack (SW.9–11, separate compose)

File	Role
compose.observe.yaml	Prometheus, Grafana, Jaeger, otel-collector
compose.otel-link.yaml	Optional: link Compose :3000 to collector for traces
instrumentation.ts + register-otel-node.ts	SW.11 traces (:8251 / optional :3000)

File	Role
docs/CXR-OTEL-LAB-MANUAL.pdf	SV.11 trace lab (Jaeger, not logs)

Chapter 5

Port map (CXR lab)

Port	Service	Role in this manual
3000	Compose cxr-ui	Claim Studio — log source
5601	Kibana	Log search UI
9200	Elasticsearch	Log index API
8251	Rehearsal dev	Traces (SW.11), <i>not</i> default log source
16686	Jaeger	Traces UI (SW.11)
3001	Grafana	Metrics dashboards (SW.10)
9090	Prometheus	Metrics scrape (SW.9)
4318	OTel collector	Trace ingest (no browser UI)

Chapter 6

Implementation checklist (repeat on another machine)

1. Start `cxr-ui` on `:3000` (`04-compose-up.sh`).
2. Confirm label `cxr.bootcamp.logs` on the service in `compose.yaml`.
3. Start ELK (`16-elk-up.sh`); run `16-elk-smoke.sh`.
4. Generate traffic on Claim Studio (`:3000`).
5. Kibana Discover on `filebeat-*`; save screenshot to `evidence/`.
6. Optional: stop ELK; proceed syllabus **SW.14 Redis**.

Chapter 7

Troubleshooting

- **No filebeat-* index** — :3000 not up or no Claim Studio traffic; wait 30s; re-run `16-elk-smoke.sh`.
- **Kibana empty Discover** — widen time range (Last 24 hours); clear KQL; refresh.
- **Only 6 docs in a 1h window** — normal if traffic was brief; generate more :3000 loads.
- **Expected logs on :8251 dev** — not shipped unless you extend Filebeat; use :3000 for SW.12.
- **Kibana slow first boot** — allow ~60s after `16-elk-up`.

Chapter 8

Study plan linkage

Item	Maps to
Starter / post-starter electives	SW.12 logs
After SW.12 evidence	SW.14+ electives
Deferred	Phase 2 M1 C4 infill

Chapter 9

Live adjustments (Kibana path + shipping)

- `compose.yaml` label `cxr.bootcamp.logs:` `"true"` on `cxr-ui` — Filebeat autodiscover target.
- Log source is Compose **:3000** container — not rehearsal **:8251** host dev.
- Kibana first screen: **Explore on my own** — skip Add integrations / sample data.
- Path: **Stack Management** → **Data Views** → **'filebeat-*'** → **Discover** (ignore Security sidebar).
- Time range: use picker **Last 24 hours** if histogram brush zoom stuck; only right after brush.
- Separate compose `compose.elk.yaml` — does not modify Claim Studio; optional down after lab.

Chapter 10

UI evidence — Kibana Discover (:5601)

Data view **new CXR** / pattern `filebeat-*`; Filebeat fields and `cxr-ui` log rows after :3000 traffic.

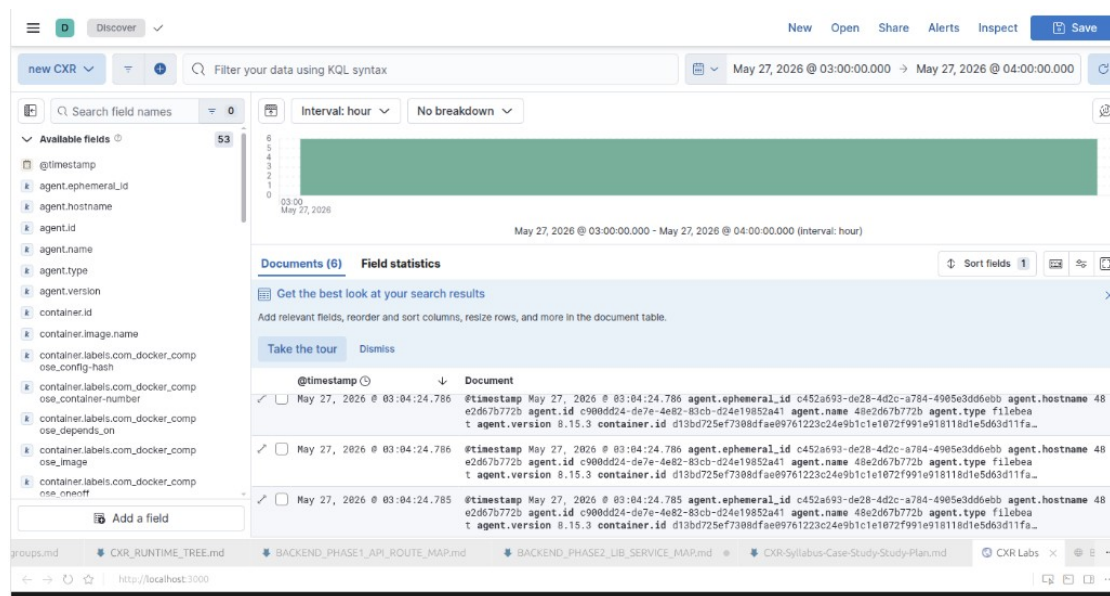


Figure 10.1: Kibana Discover — `filebeat-*` docs from :3000 `cxr-ui` (SW.12 evidence).